



GALGOTIAS COLLEGE OF ENGINEERING & TECHNOLOGY
GREATER NOIDA
DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

BTECH (SEM V)

ARTIFICIAL INTELLIGENCE (BCAI-501)

UNIT-3 ASSIGNMENT

Section A: Short Answer

1. Explain the role of quantifiers ($\forall$, $\exists$) in FOPL with examples.
 2. Write FOPL expressions for the following statements:
 - a. All humans are mortal.
 - b. Some students like mathematics.
 3. Apply the concepts of upper ontology with an example.
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Section B: Long Answer

1. Represent the information used in the mentioned problem in predicate logic. "There is a monkey at the door in a room. In the middle of the room a bunch of bananas is hanging from the ceiling. The monkey is hungry and wants to get the banana but we cannot stretch high enough from the floor. At the window of the room there is a box".
2. Convert each of the following sentences
 - first into Predicate logic
 - and then in Clausal Form/Prenex Normal Form
 - and then do Resolution for goal: - Scrooge is not a child.
 - (i) Every child loves Santa.
 - (ii) (ii) Everyone who loves Santa loves any reindeer.
 - (iii) (iii) Rudolph is a reindeer, and Rudolph has a red nose.
 - (iv) (iv) Anything which has a red nose is weird or is a clown.
 - (v) (v) No reindeer is a clown.

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